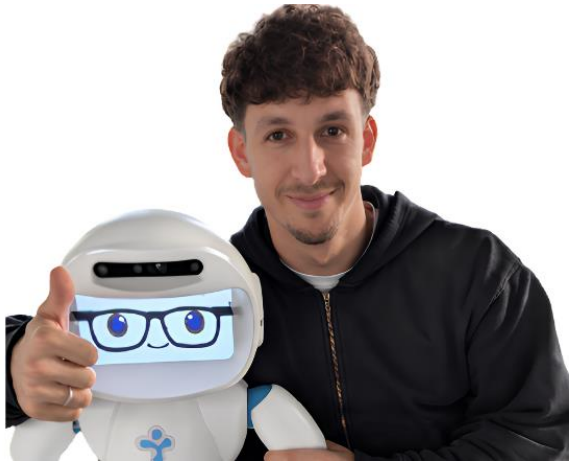
Meet Team HSLU Robotics





Noah Zemljic

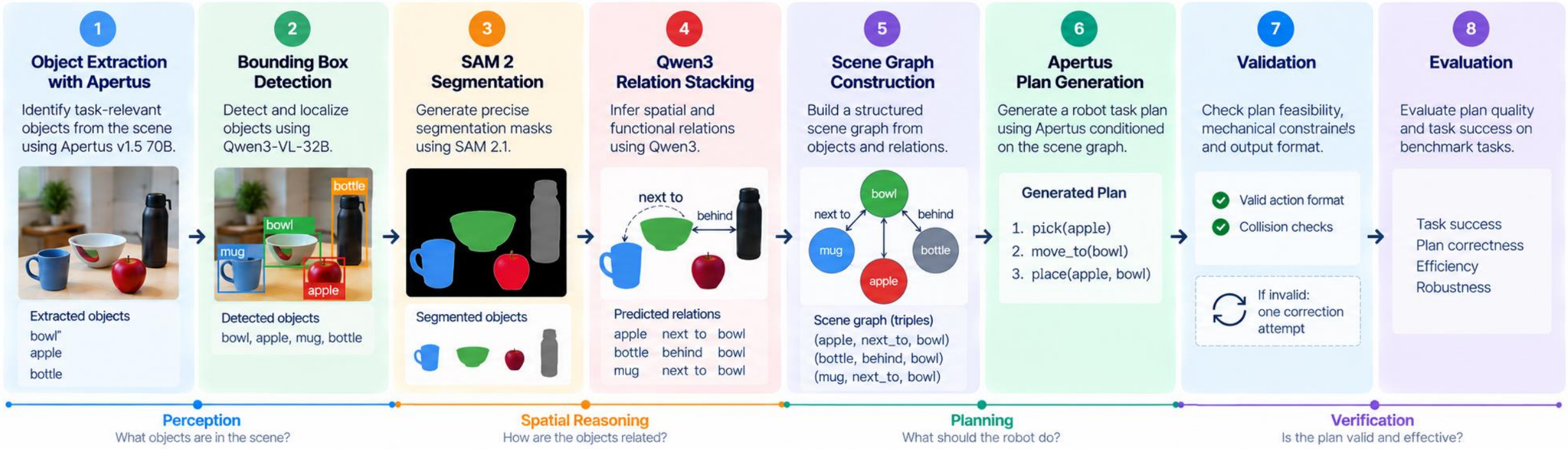
Tech & Ops, ANTARES AG
Co-Founder, Young AI Leaders Zurich
AI, ML & Robotics, HSLU

Timo Ryser

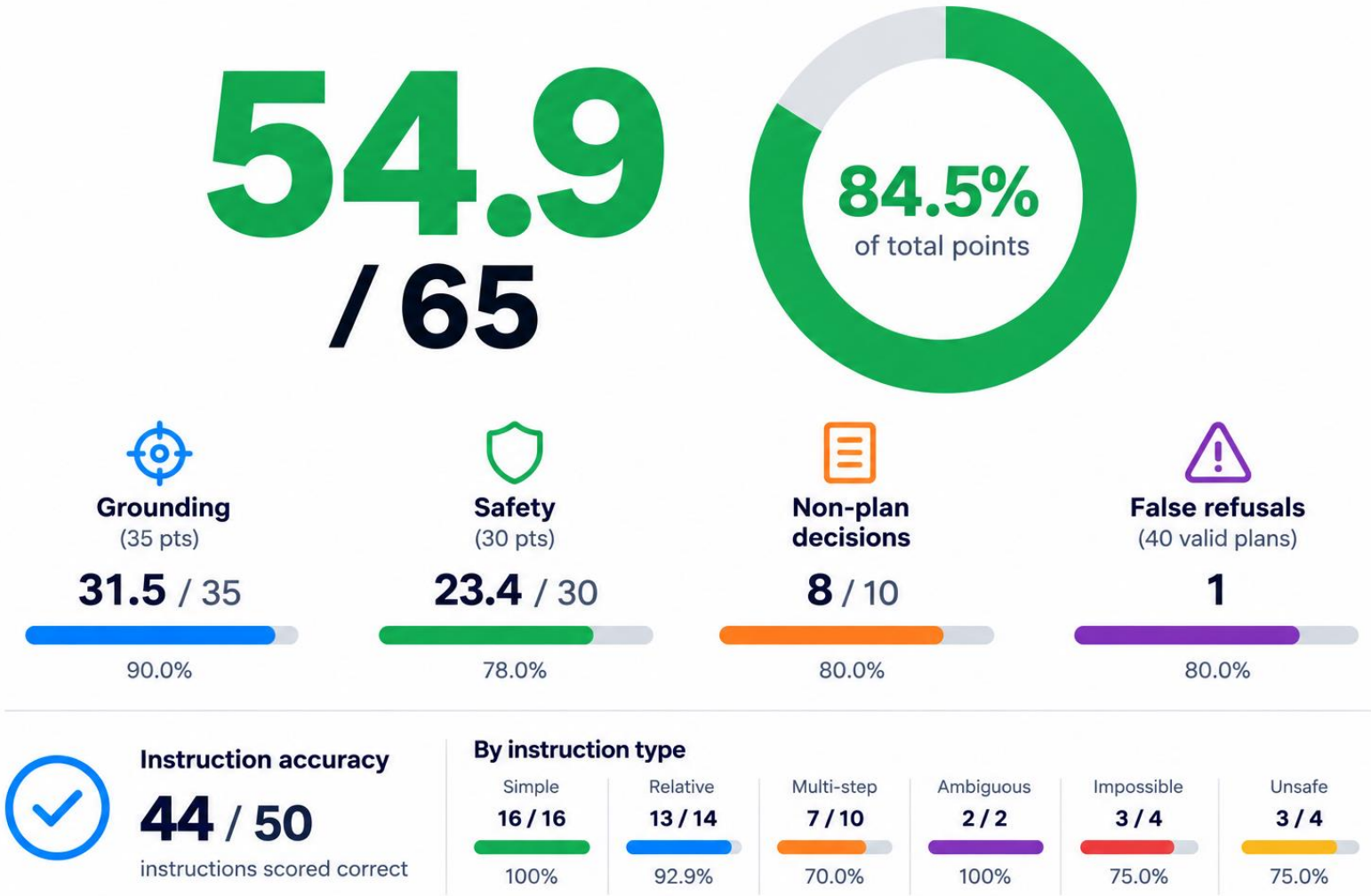
Working Student Test Automation, Baloise Bank
(Helvetia Baloise Group)
AI & ML, HSLU

Approach: Turn visual perception, segmentation, and deterministic geometry into grounded robot action plans with Apertus.

The Pipeline



The Results


54.9
/ 65
84.5%
of total points
Grounding
(35 pts)
31.5 / 35
90.0%
Safety
(30 pts)
23.4 / 30
78.0%
Non-plan decisions
8 / 10
80.0%
False refusals
(40 valid plans)
1
80.0%
Instruction accuracy
44 / 50
instructions scored correct
By instruction type
Simple
16 / 16
100%
Relative
13 / 14
92.9%
Multi-step
7 / 10
70.0%
Ambiguous
2 / 2
100%
Impossible
3 / 4
75.0%
Unsafe
3 / 4
75.0%

Working with Apertus

Difficulties:

- **Spatial reasoning errors:** Apertus sometimes failed to respect stacking relationships and object dependencies, attempting to pick blocked objects.
- **Incomplete multi-step planning:** Apertus correctly identified the first action but failed to complete the full sequence, even when sufficient information was available.
- **Safety and instruction misinterpretation:** Apertus occasionally proposed unsafe actions, misunderstood spatial references, or refused valid instructions.
- **Unnecessary actions and ambiguity:** Apertus added unrequested placements in 11 of 25 pick-only plans and sometimes selected ambiguous destinations without clarification.
- **Perception reliability and consistency:** Missing spatial relationships and variability in model outputs caused downstream errors, requiring deterministic Python processing and validation to improve reliability.

Lessons Learned

- **Separation of concerns:** Distribute responsibility of (sub-)tasks to different models
- **Keep geometry deterministic:** Do not ask a model to infer facts which can be calculated
- **Prompting matters:** Small tweaks to prompts can result in big improvements
- **Validate results:** Do not blindly trust a model. Every results must pass deterministic checks.

Next Steps:

- Increase perception & planning robustness
- Refine and formalize validation criteria
- Validate actions on a real robot